

Internal Components -- Guided Notes

Unit tp-2.2 | CompTIA Network+ N10-009 Obj. FC0-U71 2.2 | N10-008 FC0-U71 2.2

Name: _____ Date: _____ Period: _____

Complete each question using the lesson reading. Write in the blank or answer in the space provided.

Question 1

The _____ is the main circuit board that connects all other components. It holds the _____ socket (where the processor plugs in), RAM slots, PCIe slots, and SATA connectors. The chipset on the motherboard manages _____ between the CPU, RAM, and expansion cards.

Question 2

RAM stands for _____ and is a _____ (volatile/non-volatile) form of memory, meaning data is lost when power is removed. The CPU uses RAM to store data _____ is currently working on. More RAM allows more programs to run _____ without performance degradation.

Question 3

The GPU (Graphics Processing Unit) is responsible for rendering _____ on a display. It can be _____ (built into the motherboard or CPU) or _____ (a dedicated add-in card in a PCIe slot). Professional workloads like video editing and machine learning also leverage GPU _____ processing power.

Question 4

The PSU (Power Supply Unit) converts _____ current (from the wall outlet) to _____ current at voltages the computer components require (typically 3.3V, 5V, and 12V). A PSU is rated in _____, which is the maximum total power it can provide to all components at once.

Question 5

NIC stands for _____ and provides the hardware interface between the computer and the network. Wired NICs use an _____ port (RJ-45 connector). Wireless NICs use _____ or _____ radio standards. Most modern motherboards have a NIC _____ (built in).

Question 6

The CPU cooler sits on top of the CPU and dissipates _____ generated by the processor. It transfers heat from the IHS (Integrated Heat Spreader) to either a _____ (passive metal fin array) or an _____ (liquid loop). Thermal paste between the CPU and cooler improves _____ transfer.

Question 7

PCIe (Peripheral Component Interconnect Express) slots on the motherboard allow expansion cards to be added. Common expansion cards installed in PCIe slots include: _____, _____, _____, and _____ (list at least three).

Question 8 -- Real World Application

REAL WORLD: A video editor complains their computer is running very slowly when exporting 4K video, even though the system has a fast CPU. Upon investigation, the technician finds only 8 GB of RAM and a motherboard with an integrated GPU. Identify two specific component upgrades that would most improve export performance and explain why each helps.

Download the answer key from the class server:

```
wget http://[SERVER]/resources/network-engineer/tp-2-2-internal-components/tp-2-2-internal-components-answer-key.pdf
```